

Total Solution Include PD for Bidirectional Fast Charge Power Bank

1. General Description

The SW6208 is a highly integrated power management IC for fast charge power bank application, and supports A+A+B+C+L five ports. It integrates 5A switching charger, 22.5W synchronous boost, PPS/PD/QC/AFC/FCP/SCP/PE/SFCP fast charge protocol, fuel gauge, segment/led driver and power controller. With simple external components, The SW6208 provides a turn-key high efficiency solution for fast charge battery management.

2. Applications

- Power Bank
- Battery Powered Device

3. Features

• Switching Charger

- Current up to 5A , Efficiency up to 96%
- Support 4.2/4.35/4.4/4.5V Battery Voltage
- Support JEITA
- Thermal Regulation

• Synchronous Boost

- Power up to 22.5W, Efficiency up to 95%
- Load Insert Detect and Light Load Detect
- Support Wireless Charge Mode
- Support Small Current Charge Mode

• Output Fast Charge Protocol

- Support PPS/PD3.0/PD2.0
- Support QC4+/QC4/QC3.0/QC2.0
- Support AFC
- Support FCP
- Support SCP
- Support PE2.0/PE1.1
- Support SFCP

• Input Fast Charge Protocol

- Support PD3.0/PD2.0
- Support AFC
- Support FCP
- Support SCP
- Support PE1.1

• Type-C Interface

- Support Type-C Specification
- Support try.SRC Role

• BC1.2 Module

- Support BC1.2 DCP
- Support Apple & Samsung Device

• Lightning Decryption

- Support Lightning Decryption

• Fuel Gauge

- Include 12bit ADC
- Support Coulometer
- Support Segment Driver
- Support 3~5 LEDs Driver

• Fast Charge LED

- Support Fast Charge LED Driver

• WLED Driver

- Support WLED Driver

• Key Support

- Support Push Key

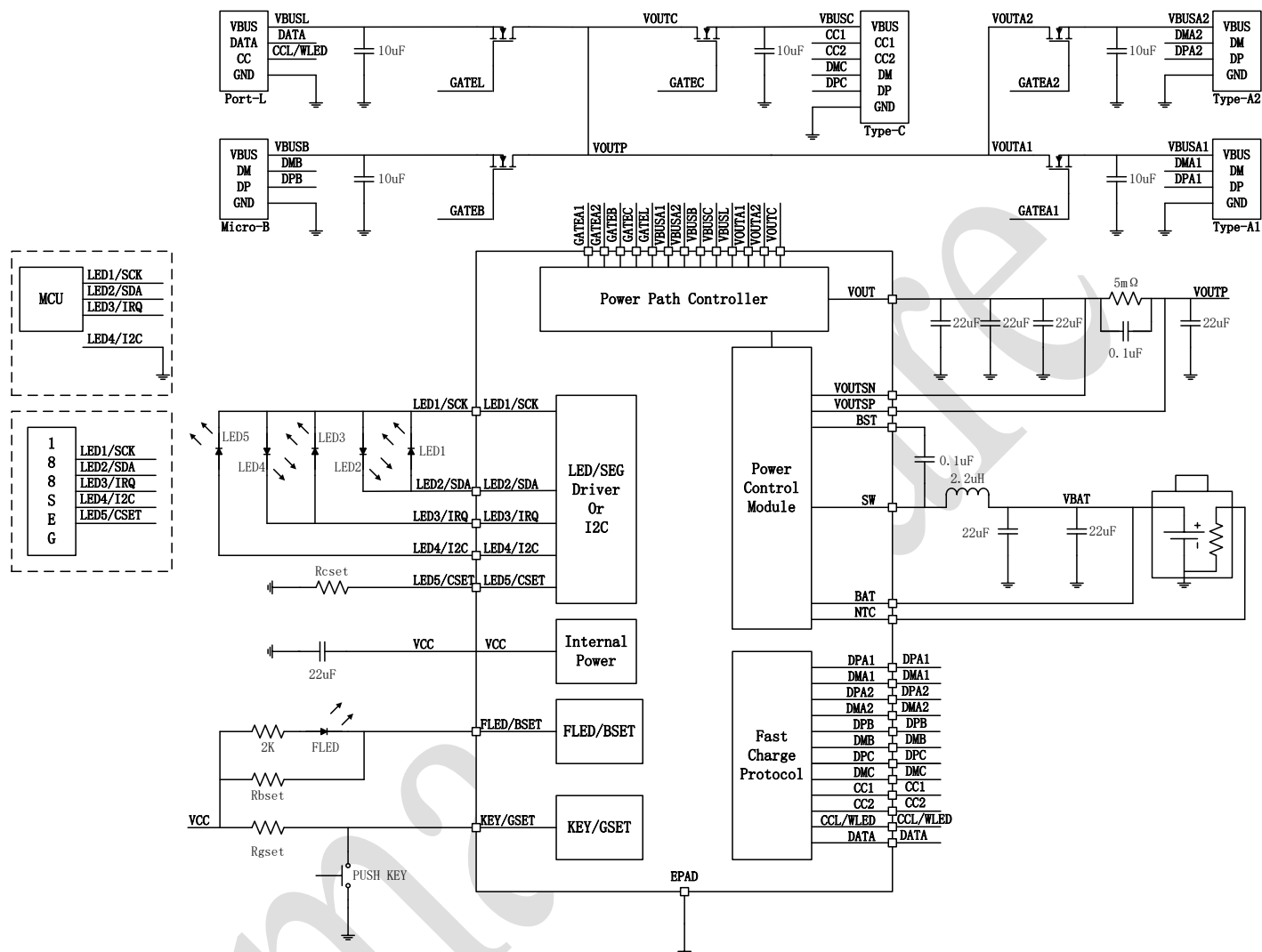
• Protection

- Output Over Current Protection
- Output Short Protection
- Charger Over Voltage Protection
- Over Temperature Protection

• I2C Interface

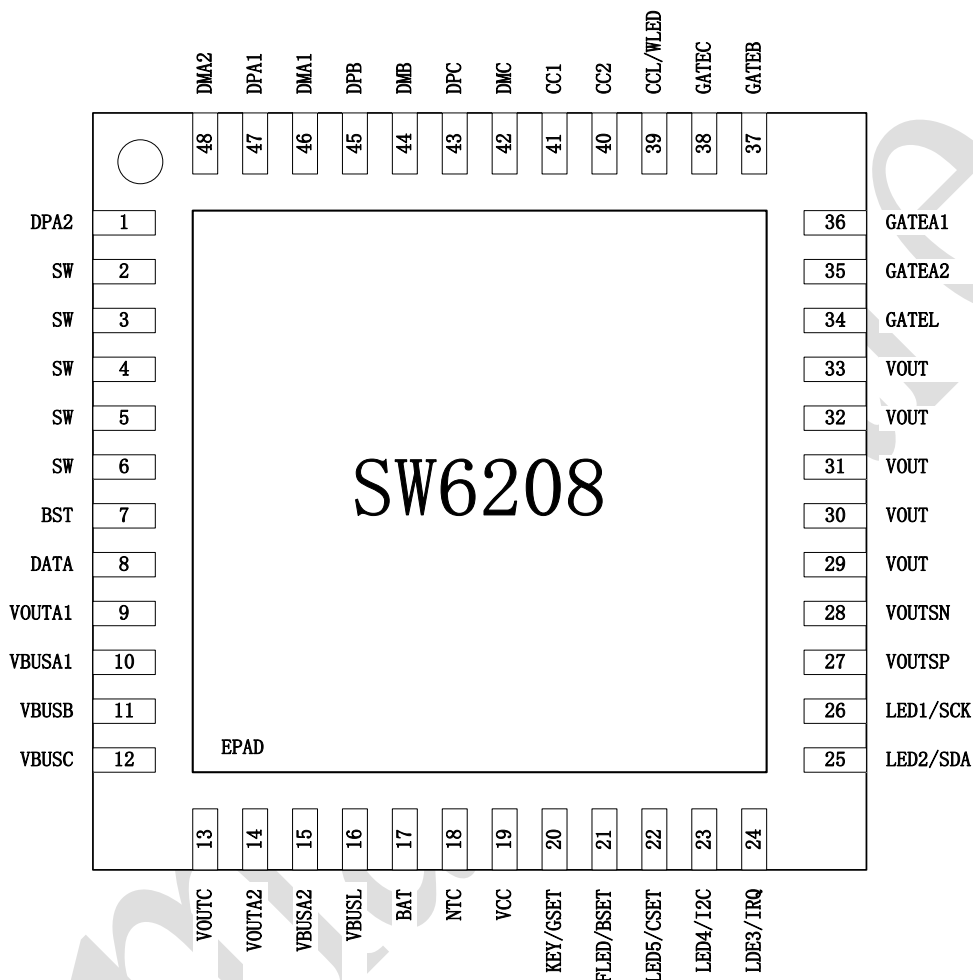
• QFN-48(6x6mm) Package

4. Functional Block Diagram



5. Pin Configuration and Function

5.1. Pin Configuration



5.2. Pin Descriptions

Pin	Name	Function Description
1	DPA2	Type-A2 port DP pin.
2, 3, 4, 5, 6	SW	Switching node.
7	BST	Bootstrap pin for high side NMOS.
8	DATA	Lightning decryption pin.
9	VOUTA1	Type-A1 port current sense pin.
10	VBUSA1	Type-A1 port current/voltage sense pin.
11	VBUSB	Micro-B port input.

12	VBUSC	Type-C port current/voltage sense pin.
13	VOUTC	Type-C port current sense pin.
14	VOUTA2	Type-A2 port current sense pin.
15	VBUSA2	Type-A2 port current/voltage sense pin.
16	VBUSL	Lightning port input.
17	BAT	Battery voltage sense pin.
18	NTC	Negative temperature coefficient (NTC) thermistor input.
19	VCC	Internal power.
20	KEY/GSET	Push key input, constant charge time set for fuel gauge.
21	FLED/BSET	Fast charge led driver, battery voltage set.
22	LED5/CSET	Led5 for segment driver and battery capacity set.
23	LED4/I2C	Led4 for segment driver, led driver or i2c set.
24	LED3/IRQ	Led3 for segment driver, led driver or irq.
25	LED2/SDA	Led2 for segment driver, led driver or i2c data.
26	LED1/SCK	Led1 for segment driver, led driver or i2c clock.
27	VOUTSP	Current sense positive pin.
28	VOUTSN	Current sense negative pin.
29, 30, 31, 32, 33	VOUT	Boost output and charger input.
34	GATEL	Lightning port power path control pin.
35	GATEA2	Type-A2 port power path control pin.
36	GATEA1	Type-A1 port power path control pin.
37	GATEB	Micro-B port power path control pin.
38	GATEC	Type-C port power path control pin.
39	CCL/WLED	Lightning port configure channel or flash led driver .
40	CC2	Type-C port configure channel CC2.
41	CC1	Type-C port configure channel CC1.
42	DMC	Type-C port DM pin.
43	DPC	Type-C port DP pin.
44	DMB	Micro-B port DM pin.
45	DPB	Micro-B port DP pin.
46	DMA1	Type-A1 port DM pin.
47	DPA1	Type-A1 port DP pin.
48	DMA2	Type-A2 port DM pin.
	EPAD	Exposed pad.

6. Absolute Maximum Ratings

Parameters	Symbol	MIN	MAX	UNIT
Input Voltage	VBUSB/VBUSC/VBUSL	-0.3	16	V
Output Voltage	VOUT/VOUTSP/VOUTSN/ VOUTA1/VOUTA2/VOUTC/ VBUSA1/VBUSA2/VBUSC	-0.3	16	V
SW Voltage	SW	-0.3	16	V
BST Voltage	BST-SW	-0.3	6	V
Power Path Control Voltage	GATEA1/GATEA2/GATEB/ GATEC/GATEL	-0.3	21	V
CC1/CC2/CCL Voltage	CC1/CC2/CCL	-0.3	16	
Other Pin Voltage		-0.3	6	V
Junction Temperature		-40	+150	°C
Storage Temperature Range		-60	+150	°C
ESD(HBM)		-4	+4	KV

【Notice】 Stresses beyond those listed under absolute maximum ratings may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under recommended operating conditions is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

7. Recommended Operating Conditions

Parameters	Symbol	MIN	Typical	MAX	UNIT
Input Voltage	VBUSB/VBUSC/VBUSL	4.5		13.5	V
Battery Voltage	BAT	2.8		4.5	V

8. Electrical Characteristics

($V_{IN} = 5V$, $V_{BAT} = 3.7V$, $T_A = 25^\circ C$, unless otherwise specified.)

Parameters	Symbol	Test Conditions	MIN	TYP	MAX	UNIT
Power Supply						
VBUSB/VBUSC/VBUSL Input Voltage	$V_{BUSB/C/L}$	$V_{BUSB}/V_{BUSC}/V_{BUSL}$	4		13.5	V
VBUSB/VBUSC/VBUSL Input UVLO Threshold	$V_{BUSB/C/L_UVLO}$	VBUSB/VBUSC/VBUSL Voltage Falling	3.9	4	4.1	V

VBUSB/VBUSC/VBUSL Input UVLO Hysteresis	V _{BUSB/C/L_UVLO_HYS}	VBUSB/VBUSC/VBUSL Voltage Rising	300	400	500	mV
VBUSB/VBUSC/VBUSL Input OVP Threshold	V _{BUSB/C/L_OVP}	VBUSB/VBUSC/VBUSL Voltage Rising	13.4	13.7	14	V
VBUSB/VBUSC/VBUSL Input OVP Hysteresis	V _{BUSB/C/L_OVP_HYS}	VBUSB/VBUSC/VBUSL Voltage Falling	600	800	1000	mV
VCC Output Voltage	V _{CC}	Boost or V _{BUSB/VBUSC/VBUSL} Insert	4.9	5	5.1	V
		Power Off		V _{BAT}		V
VCC Output current	I _{CC}	Boost or V _{BUSB/VBUSC/VBUSL} Insert	40	60	80	mA
		Power Off	40	60	80	mA
Power MOS Rdson						
High Side NMOS	R _{DSON_H}		17	20	24	mΩ
Low Side NMOS	R _{DSON_L}		9	11	14	mΩ
Peak Current Limit for High Side NMOS	I _{PEAK_H}	Charge Mode	6	8	10	A
Peak Current Limit for Low Side NMOS	I _{PEAK_L}	Boost Mode	8	10	12	A
Charge Mode						
Trickle Charge End Voltage	V _{TC}		2.9	3	3.1	V
Trickle Charge Current(IBAT)	I _{TC}	V _{BAT} < 0.5V	30	60	100	mA
		0.5V < V _{BAT} < 3V	200	300	400	mA
Constant Current Charge Current	I _{CC}	V _{BUSB} / V _{BUSL} = 5V, I _{CCSET} =2A	1.8	2	2.1	A
		V _{BUSC} = 5V, I _{CCSET} =3A	2.7	3	3.2	A
		V _{BUSB} / V _{BUSC} / V _{BUSL} = 9V, I _{CCSET} =2A	1.8	2	2.1	A
		V _{BUSB} / V _{BUSC} / V _{BUSL} =12V, I _{CCSET} =1.5A	1.3	1.5	1.6	A
Termination Charge Current	I _{END}	V _{BUSB} / V _{BUSC} / V _{BUSL} = 5V	200	230	260	mA
		V _{BUSB} / V _{BUSC} / V _{BUSL} = 9V	100	130	160	mA
		V _{BUSB} / V _{BUSC} / V _{BUSL} = 12V	80	100	120	mA
Target Charge Voltage	V _{BAT_FULL}		4.16	4.2	4.24	V
Recharge Threshold	V _{BAT_RECH}		4.06	4.1	4.14	V

Switching Frequency	F _{CHG}		350	400	450	KHz
Trickle Charge Over Time	t _{TC_OT}		36	40	44	Min
Constant Charge Over Time	t _{CC_OT}		30	33	36	Hour
Thermal Regulation Threshold	T _{REGU_CHG}		100	115	130	℃
Hold Voltage Threshold	V _{HOLD}	V _{BUSB} / V _{BUSC} / V _{BUSL} = 5V	4.4	4.5	4.6	V
		V _{BUSB} / V _{BUSC} / V _{BUSL} = 9V	8.4	8.5	8.6	V
		V _{BUSB} / V _{BUSC} / V _{BUSL} = 12V	11.35	11.45	11.55	V
Boost Mode						
VBAT Input Voltage	V _{BAT}		2.9		4.5	V
VBAT Input UVLO Threshold	V _{BAT_UVLO}	V _{BAT} Voltage Falling	2.8	2.9	3.0	V
VBAT Input UVLO Hysteresis	V _{BAT_UVLO_HYS}	V _{BAT} Voltage Rising	400	500	600	mV
VOUT Output Voltage	V _{OUT}	V _{OUT} =5V, I _{OUT} =0A	5	5.1	5.2	V
		V _{OUT} =9V, I _{OUT} =0A	8.9	9.1	9.3	V
		V _{OUT} =12V, I _{OUT} =0A	11.8	12.1	12.4	V
VOUT Output Current	I _{OUT}	V _{OUT} =5V, I _{OUTSET} =3A	3		3.4	A
		V _{OUT} =9V, I _{OUTSET} =2A	2		2.3	A
		V _{OUT} =12V, I _{OUTSET} =1.5A	1.5		1.8	A
Light Load Current	I _{LIGHT_LOAD}	R _{DS_PATH} =10mΩ , V _{OUT} =5V	40	60	80	mA
		R _{DS_PATH} =10mΩ , V _{OUT} =9V/12V	20	40	60	mA
Light Load Time	t _{LIGHT_LOAD}	Single Port	28	32	40	S
		Multiple Ports	12	16	20	S
Quiescent Current	I _Q	V _{BAT} =3.7V	40	50	65	uA
Wire Drop Compensation	V _{OUT_WDC}	0A<I _{OUT} <1A		0		mV
		1A<I _{OUT} <2A	30	50	70	mV
		I _{OUT} >2A	80	100	120	mV
Switching Frequency	F _{BST}		350	400	450	KHz
Thermal Regulation Threshold	T _{REGU_BST}		100	115	130	℃
Type-C						
CC Current Source	I _{CC_SOURCE}	Power Level=3.0A	310	330	350	uA
CC Termination Resistor	R _D		4.9	5.1	5.3	kΩ
BC1.2						

DP/DM Voltage	DP	Apple 2.4A Mode	2.55	2.7	2.85	V
	DM	Apple 2.4A Mode	2.55	2.7	2.85	V
PE						
Current Threshold	I _{REF}		150	250	350	mA
Quit Time	t _{PLUG_OUT}		160	200	240	mS
LED						
Segment Source Current	I _{SEG}		6	10	14	mA
LED Source Current	I _{LED}		2	4	6	mA
LED Flicker Frequency	f _{LED}		0.8	1	1.2	Hz
WLED						
WLED Resistor	R _{WLED}		10	20	30	Ω
KEY						
Short Key Time	T _{SHORT}		24	32	500	mS
Long Key Time	T _{LONG}		1.5	2	3	S
Double Short Key	T _{DOUBLE}	Longest Time For Two Short Key Happen	1.2	1.5	1.8	S
I2C						
Rate	f _{CLK}			100	400	Kbit/S
Thermal Shutdown						
Thermal Shutdown Threshold	T _{SHDT}	Temperature Rising	135	150	165	℃
Thermal Shutdown Hysteresis	T _{SHDT_HYS}	Temperature Falling	55	70	85	℃

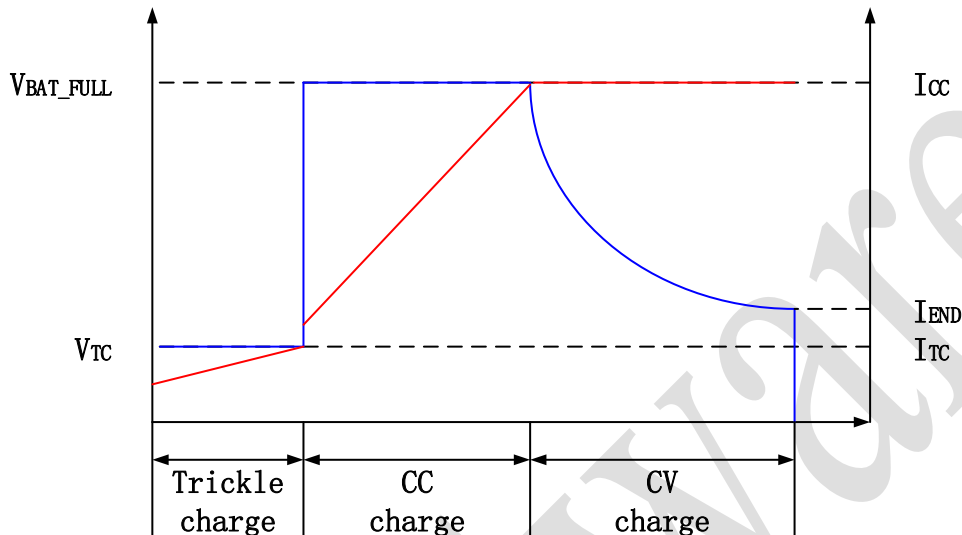
9. Functional Description

9.1. Charge Mode

The SW6208 integrates a switching charger with efficiency is up to 96%. The switching charger supports 4.2V/4.35V/4.4V/4.5V battery voltage. Its switching frequency is up to 400KHz, so a small size inductor with inductance of 2.2uH can be used.

The switching charger charges battery in three charge phases: trickle charge(TC), constant current charge(CC) and constant voltage charge(CV). When battery voltage is lower than 3V, charger enters into trickle charge while the charge current is small current. When battery voltage rises to 3V, charge enters into constant current charge while charge current is fast charge current. When battery voltage

risers to target charge voltage, charger enters into constant voltage charge while charge current falls so as battery voltage stays to target voltage. When charge current falls to termination charge current, charge cycle completes and charger stops. When battery voltage is lower than recharge threshold, charger automatically restarts.



The battery target voltage can be set by FLED/BSET pin. When FLED/BSET is pulled up to VCC with a 10K Ω resistor, 4.2V is set; when selecting a 15K Ω resistor, 4.35V is set; when selecting a 5.6K Ω resistor, 4.4V is set; when selecting a 3K Ω resistor, 4.5V is set.

The charge current in input current varies with input voltage, while battery current is up to 5A. When input voltage is 5V, input charge current is set to 2A for Micro-B and Lightning port, and 3A for Type-C port. When input voltage is 9V, input charge current is set to 2A. When input voltage is 12V, input charge current is set to 1.5A.

The switching charger has battery temperature protection and support JEITA. It monitors battery temperature by measuring the voltage of NTC pin which connects to a negative temperature coefficient (NTC) thermistor. When battery temperature is abnormal, charger will reduce charge current, reduce target charge voltage or stop to protect battery. While using a typical NTC thermistor of 103AT, When battery temperature is lower than 5°C, charger will reduce charge current by half; When battery temperature continues to fall to lower than 0°C, charger will stop; When battery temperature rises to 5°C, charger will restart and reduce charge current by half; When battery temperature continues to rise to higher than 10°C, charge current will return to normal; When battery temperature is higher than 45°C, charger will reduce target charge voltage by 0.1V; When battery temperature continues to rise to higher than 50°C, charger will stop; When battery temperature falls to 45°C, charger will restart and reduce target charge voltage by 0.1V; When battery temperature continues to fall to lower than 40°C, target charge voltage will return to normal. Resistors can be in series or parallel with the NTC thermistor to change the protection temperature range. JEITA function can be disabled if not needing.

The switching charger integrates thermal regulation. When die temperature rises to the charge thermal regulation threshold, charge current will automatically fall down so the die temperature stops increasing. If temperature continues rising to thermal shutdown temperature, charger will stop and the

SW6208 powers off.

The switching charger integrates over time protection. When charging remains in trickle charge beyond the trickle charge over time, t_{TC_OT} , or charging remains in constant current charge beyond the constant current charge over time, t_{CC_OT} , charger will terminate. Charger will restart only when re-inserting the adapter.

9.2. Boost Mode

The SW6208 integrates a synchronous boost with output power up to 22.5W and efficiency is up to 95%.

The synchronous boost works in PSM/PWM mode. It works in PSM mode when in light load and in PWM mode in heavy load to make a better efficiency. It will automatically change in these two modes base on output current.

The synchronous boost supports CC/CV mode. When output current is lower than CC limited current, output voltage will keep constant. When output current reaches CC limited current, output voltage will drop to keep output current constant. The CC limited current value is set according to the output voltage. The higher the output voltage, the lower CC limited current value.

The synchronous boost has battery temperature protection. When battery temperature is abnormal, boost will stop to protect battery. While using a typical NTC thermistor of 103AT, When battery temperature is higher than 60°C or lower than -20°C, boost will stop and turn off. When battery temperature enters into normal range, boost will not automatically restart except a start condition such as short key or load detect occur. Resistors can be in series or parallel with the NTC thermistor to change the protection temperature range.

The synchronous boost integrates thermal regulation. When die temperature rises to boost thermal regulation threshold, output voltage will automatically fall down so as die temperature stop increasing. If die temperature still rises to thermal shutdown threshold, boost will stop and turn off. When die temperature falls to thermal shutdown hysteresis, boost will not automatically restart except a start condition such as short key or load detect occur.

The synchronous boost integrates input under voltage, output over voltage, output overload and short protection.

9.3. Power Path

The SW6208 supports Type-A1+Type-A2+Micro-B+Type-C+Lightning. Type-A1 and Type-A2 port supports QC3.0/QC2.0/AFC/FCP/SCP/PE2.0/PE1.1/SFCP fast charge output. Type-C port supports PPS/PD3.0/PD2.0/

QC4+/QC3.0/QC2.0/AFC/FCP/SCP/PE2.0/PE1.1/SFCP fast charge output, and

PD3.0/PD2.0/AFC/FCP/SCP/

PE1.1 fast charge input. Micro-B supports AFC/FCP/SCP/PE1.1 fast charge input. Lightning supports PD3.0/PD2.0 fast charge input, and Lightning cable decryption.

When short key occurs or load inserts, Type-A1/Type-A2 port will turn on to power supply extern device, and light load detection will turn off Type-A1/Type-A2/Type-C port. Light load current is related to power path MOS. if power path MOS $R_{ds(on)}$ is $10m\Omega$, the light load current will be about 60mA/40mA when output voltage is 5V/high voltage. When SOURCE is attached, Type-C port will turn on and charger will automatically turn on to charge battery. When SINK is attached, Type-C port will turn on and boost will automatically turn on to supply device. When SINK is unattached, Type-C port will turn off and boost will automatically turn off. Type-C port also support light load detection. When SINK is light load, Type-C port and boost will turn off to reduce power consumption. When adapter inserts, Micro-B or Lightning port will turn on and charger will automatically turn on.

The SW6208 supports charging the battery and supplying extern device simultaneously, and this function can be disabled if not needing. When only one port turns on, it supports fast charge input and output. When two or three ports turn on, it only supports 5V input and output. In this situation, charger hold voltage is set to 4.8V. if input voltage drops due to the weak power supply of adapter, charge current will automatically fall down so the input voltage holds to 4.8V and supplies extern device in priority.

The SW6208 supports Type-A1/Type-A2/Type-C ports supplying extern devices simultaneously and only supports 5V output.

9.4. Mode Set

The SW6208 supports wireless charge, small current charge and flash light driver modes. These three modes are set by register in I2C mode, and set by pin in segment/led mode.

The SW6208 supports wireless charge mode. In wireless charge mode, wireless charge module is connected to Type-A2 port. Light load threshold of Type-A2 port is set to 240mA/115mA in 5V/high voltage output, and light load detect time is set to 32S. Wireless charge mode is set when LED1/SCK pin connects a $100K\Omega$ resister to ground.

The SW6208 supports small current charge mode, and can charge small current device such as bluetooth headset. When enter into small current charge mode, light load detect will be disabled for two hours, and double short key will also quit this mode. Besides, fuel gauge display will also change to indicate it is in small current charge mode. Small current charge mode is set when LED2/SDA pin connects a $100K\Omega$ resister to ground.

The SW6208 supports flash light driver mode also called WLED mode. CCL/WLED Pin can be configured to WLED driver or PD fast input function of Lightning port, only one of these two functions can be used in one application. When LED3/IRQ pin connects a $100K\Omega$ resister to ground, CCL/WLED pin is configured to WLED driver, or configured to PD fast input function of Lightning port.

9.5. Type C Interface

The SW6208 integrates Type-C logic controller include try.SRC role. When SOURCE is attached, charger will automatically turn on to charge battery. When SOURCE is detached, charger will automatically turn off. When SINK is attached, boost will automatically turn on to supply device. When SINK is detached, boost will turn off.

When SINK is attached and boost turns on, the SW6208 works in SOURCE role, and will broadcast power level of 3A. if adapter inserts in Micro-B or Lightning port and boost not turns on, the SW6208 will also broadcast power level of 3A.

9.6. PD Fast Charge

The SW6208 integrates PPS/PD3.0/PD2.0 fast charge protocol, supports input and output fast charge. Its output supports two APDOs: PPS0 for 5~5.9V@3A, and PPS1 for 5~11V@2A, and three fix PDOs: 5V@3A/2A/2.4A/2.5A, 9V@2A/2.22A/2.33A/2.4A, 12V@1.5A/1.6A/1.67A/1.75A. It also supports 5V/9V/12V input voltage.

9.7. QC3.0/QC2.0 Fast Charge

The SW6208 integrates QC4+/QC4/QC3.0/QC2.0 fast charge protocol. It supports Class A, while QC2.0 supporting 5V/9V/12V output voltage and QC3.0 supporting 5V~12V output voltage, 200mV/Step.

QC2.0/QC3.0 will output voltage base on DP/DM voltage:

Device		SW6208	
DP	DM	VOUT	Note
3.3V	3.3V	previous voltage	not respond
0.6V	0.6V	12V	
3.3V	0.6V	9V	
0.6V	3.3V	continuous mode	0.2V/Step

0.6V	GND	5V	
------	-----	----	--

9.8. AFC Fast Charge

The SW6208 integrates AFC fast charge protocol, supports 5V/9V/12V output voltage and 5V/9V input voltage.

9.9. FCP Fast Charge

The SW6208 integrates FCP fast charge protocol, supports 5V/9V/12V output voltage and 5V/9V input voltage.

9.10. SCP Fast Charge

The SW6208 integrates SCP fast charge protocol, supports 5V@4.5A output voltage and 5.5V/3A input voltage.

9.11. PE Fast Charge

The SW6208 integrates PE2.0/PE1.1 fast charge protocol. PE2.0 supports 5V~12V output voltage, 500mV/Step. PE1.1 supports 5V/7V/9V/12V output voltage and 5V/9V input voltage.

9.12. SFCP Fast Charge

The SW6208 integrates SFCP fast charge protocol, supports 5V/9V/12V output voltage.

9.13. BC1.2 Module

The SW6208 integrates BC1.2 controller, and automatically detects apple and samsung devices:

Apple 2.4A mode: DP=2.7V, DM=2.7V;

Samsung 2A mode: DP=1.2V, DM=1.2V;

9.14. Lightning Decryption

The SW6208 integrates lightning decryption and supports inner decryption or external decryption. When selecting inner decryption, DATA pin is directly connected to Lightning port. When selecting external decryption, DATA pin is connected to VCC, and use external decryption chip to decrypt Lightning cable.

9.15. Fuel Gauge

The SW6208 integrates coulometer to obtain accurate battery capacity.

The coulometer supports battery maximal capacity self-learning, and can learn the battery current maximal capacity in one complete charge cycle .

The battery initial capacity can be set by a resister which connecting LED5/CSET pin to ground. The relation between initial capacity Cset and resister Rset is:

$$Rset = (Cset + 2000) * 5/3$$

While Cset unit is mAh, and Rset unit is Ω .

The constant charge time can be set by a resister which connecting KEY/GSET pin to VCC.

9.16. ADC

The SW6208 integrates 12 bit ADC, and samples VOUT voltage/IOUT current/battery voltage/NTC voltage/chip temperature.

ADC channel	Range	Step
VOUT voltage	0~16.384V	4mV
IOUT current	0~9.309A	25/11mA
Battery voltage	0~4.915V	1.2mV
NTC voltage	0~4.505V	1.1mV
Chip temperature	-100~200℃	1/6.82℃

9.17. NTC Function

The SW6208 integrates NTC function. It monitors battery temperature and protects when battery temperature is abnormal. NTC Pin will source a current to a NTC thermistor of 103AT, and measure the voltage to calculate battery temperature. NTC Pin will source 80uA for higher accuracy when NTC thermistor is in low resistance, and 40uA for wider range in high resistance. It sources 80uA in default mode. When NTC voltage is higher than 2.712V, it changes to 40uA; When NTC voltage is lower than 0.718V, it changes to 80uA.

While using a typical NTC thermistor of 103AT, protection threshold and NTC voltage in discharge mode is as below:

Threshold Description	NTC Temperature/°C	NTC Voltage/V	NTC Current/uA
NTC Low Temperature Protection	-20	2.711	40
NTC High Temperature Regulation	55	0.283	80
NTC High Temperature Protection	60	0.242	80

Protection threshold and NTC voltage in Charge mode is as below:

Threshold Description	NTC Temperature/°C	NTC Voltage/V	NTC Current/uA
NTC Low Temperature Protection	0	2.182	80
NTC JEITA Low Temperature for Reducing Current	5	1.764	80
NTC JEITA Low Temperature for Returning to Normal	10	1.437	80
NTC JEITA High Temperature for Returning to Normal	40	0.466	80
NTC JEITA High Temperature for Reducing Target Voltage	45	0.393	80
NTC High Temperature Protection	50	0.333	80

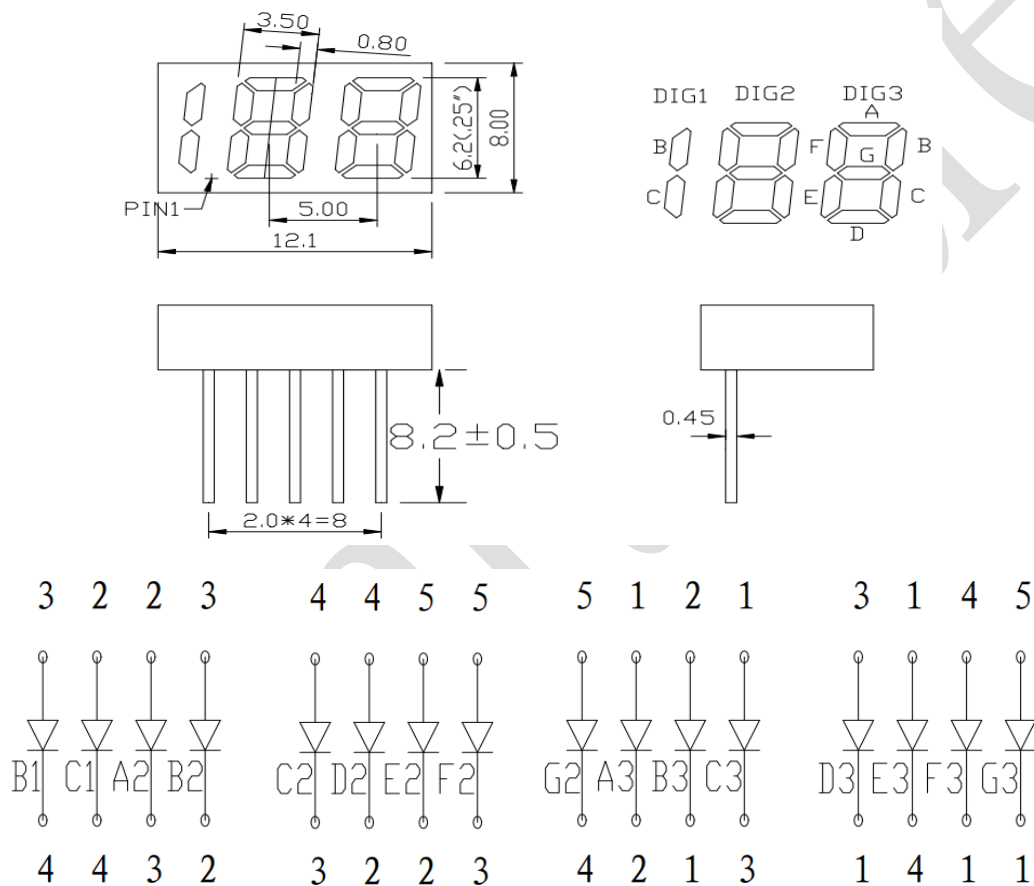
Resisters can be in series or parallel with the NTC thermistor to change the protection temperature

range. If not needing NTC protection, connect NTC pin to GND.

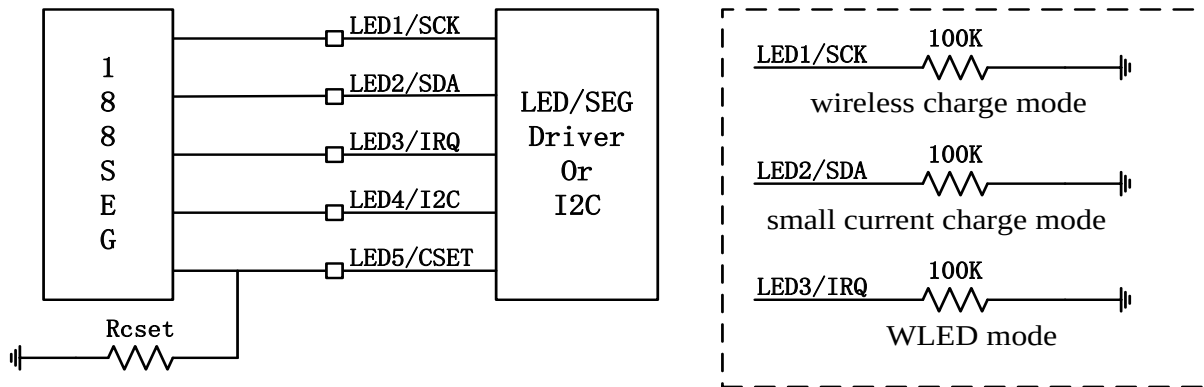
9.18. Segment Driver

The SW6208 integrates segment driver, and supports normal 188 segment and fast charge 188 segment with % and fast charge indication.

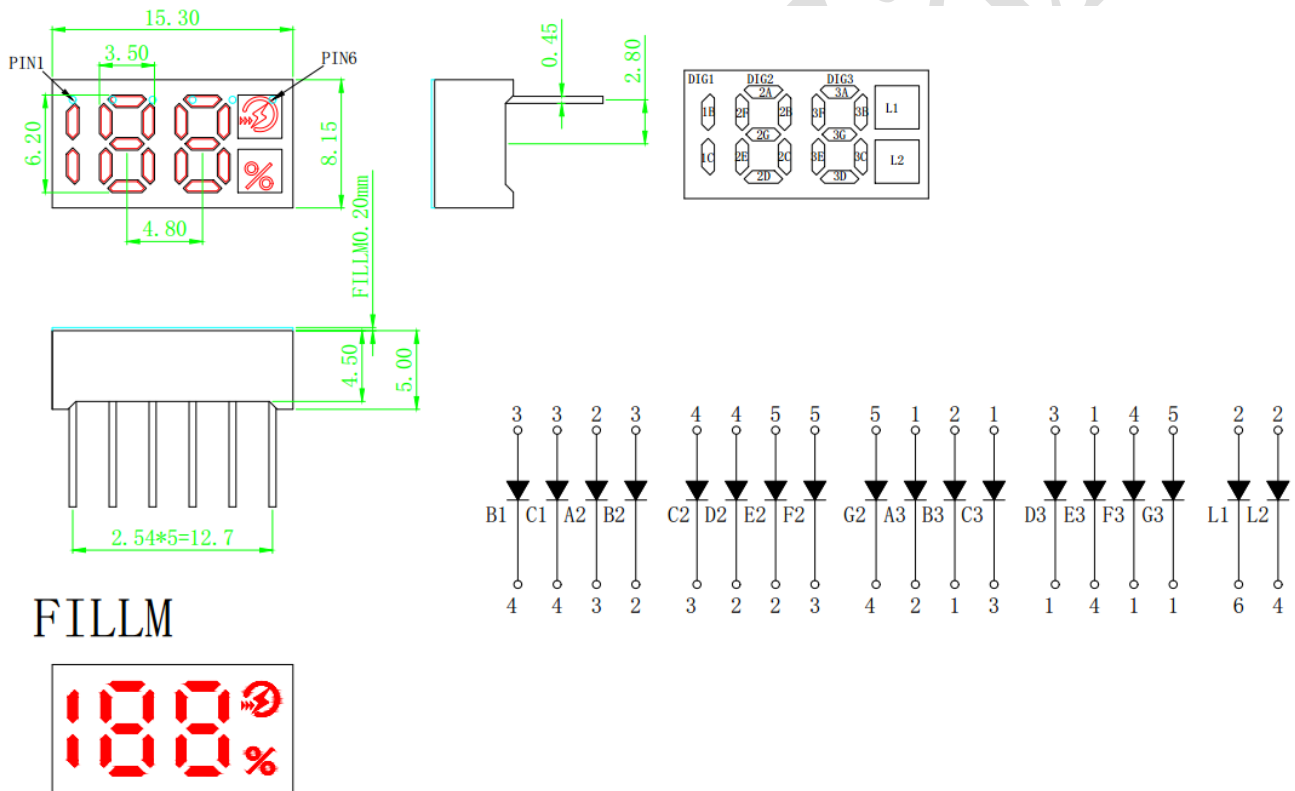
The normal 188 segment type is YF2252S-5. Its mechanical outline and circuit diagram is as below:



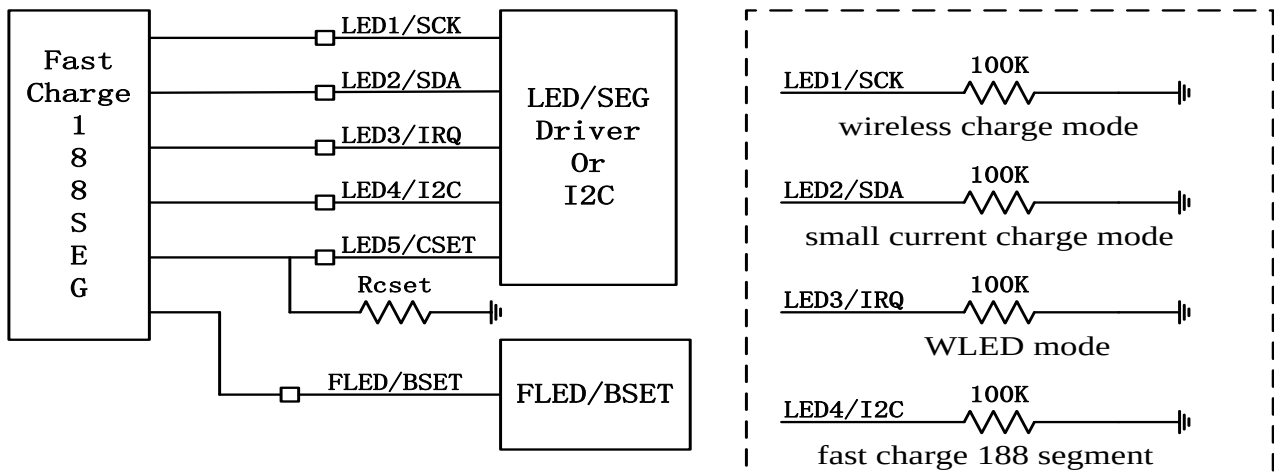
The normal 188 segment driver connection is as below:



The fast charge 188 segment type is KHP302518, and is set when LED4/I2C pin connects a 100K Ω resistor to ground. Its mechanical outline and circuit diagram is as below:



The fast charge 188 segment driver connection is as below:



In discharge mode, segment will keep on to indicate battery capacity. When battery capacity is lower than 5%, segment will flicker with frequency of 1Hz. When battery is low power, segment 0% will flicker 5 times to indicate battery needs to charge and then power off.

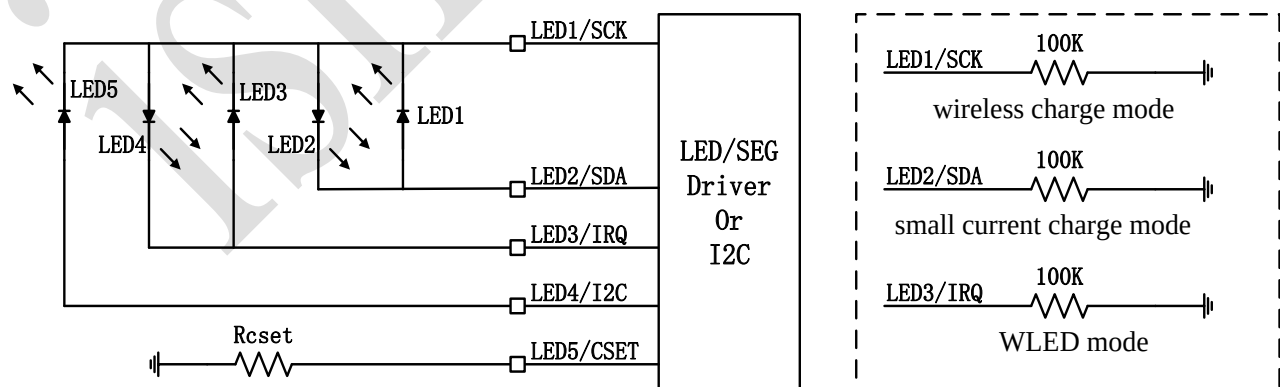
In charge mode, segment units will flicker to indicate battery capacity.

In abnormal situation such as output over current, output short, input over voltage, chip over temperature and NTC protection, segment will flicker 5 times to indicate abnormal situation and then power off.

9.19. Led Driver

The led driver supports 3/4/5 LEDs.

When connecting 5 LEDs, the LEDs connect way is as below:



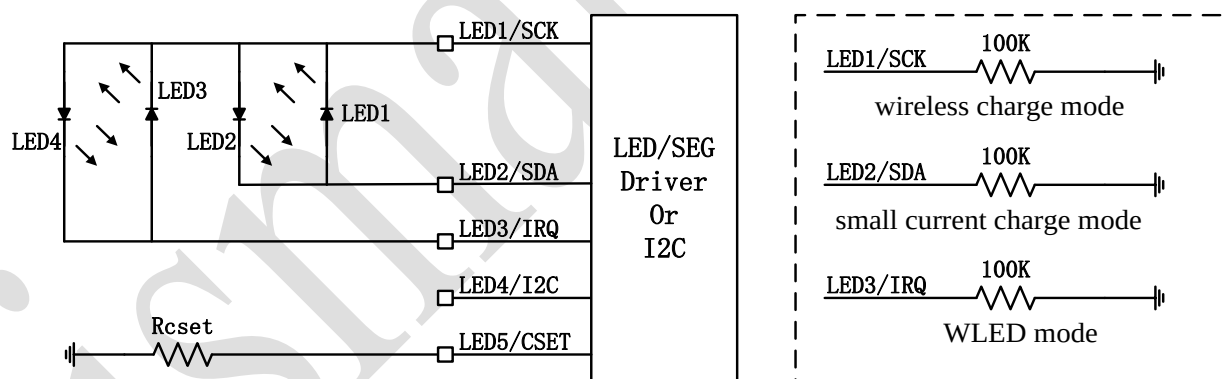
When discharging, battery capacity is shown as:

Capacity	LED1	LED2	LED3	LED4	LED5
80~100%	On	On	On	On	On
60~80%	On	On	On	On	Off
40~60%	On	On	On	Off	Off
20~40%	On	On	Off	Off	Off
5~20%	On	Off	Off	Off	Off
1~5%	Flicker	Off	Off	Off	Off
0%	Off	Off	Off	Off	Off

When charging, battery capacity is shown as:

Capacity	LED1	LED2	LED3	LED4	LED5
100%	On	On	On	On	On
80~99%	On	On	On	On	Flicker
60~80%	On	On	On	Flicker	Off
40~60%	On	On	Flicker	Off	Off
20~40%	On	Flicker	Off	Off	Off
0~20%	Flicker	Off	Off	Off	Off

When connecting 4 LEDs, the LEDs connect way is as below:



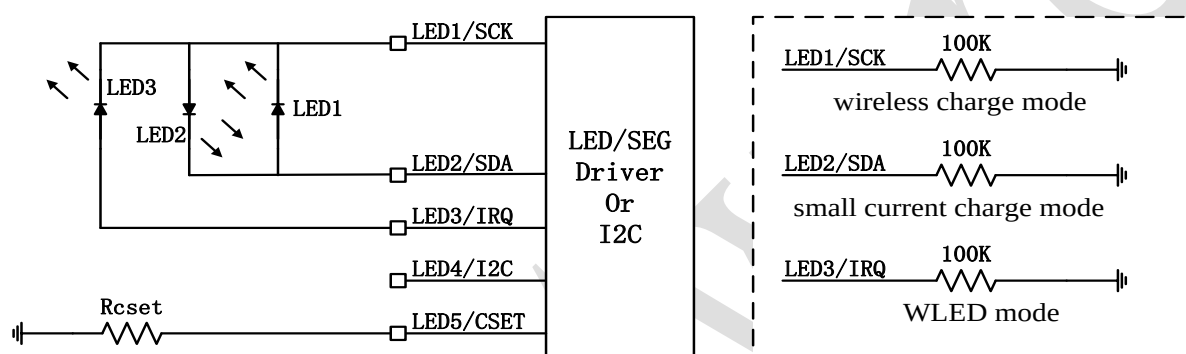
When discharging, battery capacity is shown as:

Capacity	LED1	LED2	LED3	LED4
75~100%	On	On	On	On
50~75%	On	On	On	Off
25~50%	On	On	Off	Off
5~25%	On	Off	Off	Off
1~5%	Flicker	Off	Off	Off
0%	Off	Off	Off	Off

When charging, battery capacity is shown as:

Capacity	LED1	LED2	LED3	LED4
100%	On	On	On	On
75~99%	On	On	On	Flicker
50~75%	On	On	Flicker	Off
25~50%	On	Flicker	Off	Off
0~25%	Flicker	Off	Off	Off

When connecting 3 LEDs, the LEDs connect way is as below:



When discharging, battery capacity is shown as:

Capacity	LED1	LED2	LED3
66~100%	On	On	On
33~66%	On	On	Off
5~33%	On	Off	Off
1~5%	Flicker	Off	Off
0%	Off	Off	Off

When charging, battery capacity is shown as:

Capacity	LED1	LED2	LED3
100%	On	On	On
66~99%	On	On	Flicker
33~66%	On	Flicker	Off
0~33%	Flicker	Off	Off

The led driver supports low power indication. When battery is low power, LED1 will flicker 5 times to indicate battery needs to charge and then power off.

In abnormal situation such as output over current, output short, input over voltage, chip over temperature and NTC protection, all LED will flicker 5 times to indicate abnormal situation and then power off.

9.20. WLED Driver

The SW6208 integrates flash light driver, and controlled by long key. WLED driver and PD fast input function of Lightning port is shared with CCL/WLED pin, and is set when LED3/IRQ pin connects a 100K Ω resistor to ground.

9.21. Fast Charge Led

The SW6208 integrates fast charge LED driver through FLED/BSET pin. When in input or output fast charge status, FLED/BSET pin will drive low to turn on fast charge led.

9.22. KEY

The SW6208 integrates key function. It supports push key, and supports short key, long key and double short key.

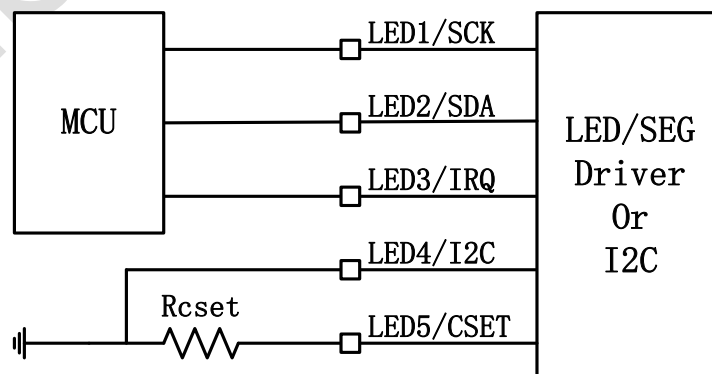
Short key: Type-A1 port, Type-C port of light load and segment/led driver will turn on;

Long key: WLED driver will turn on or turn off; When in small current charge mode, enter or quit small current charge mode.

Double short key: Type-A1/Type-A2 port, Type-C output port and segment/led driver will turn off ;

9.23. I2C Interface

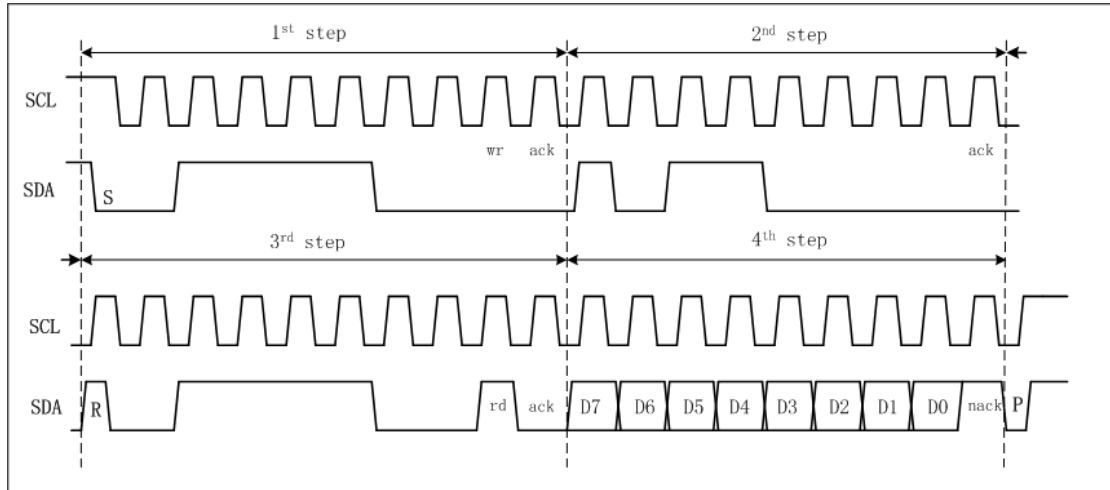
The SW6208 integrates I2C interface, supports 100K/400K rate. I2C interface and Segment/led driver are shared the same four pins, identifying by setting led4/i2c pin. When using as I2C interface, LED4/I2C should connect to ground. Wireless charge, small current charge and flash light driver modes are set by register in I2C mode.



Read Timing:

Slave address : 0x3C

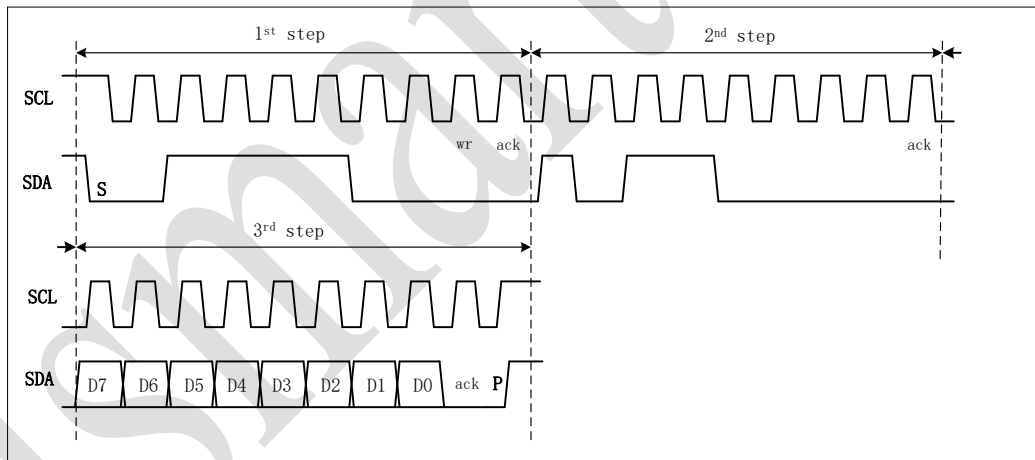
Register address: 0xB0



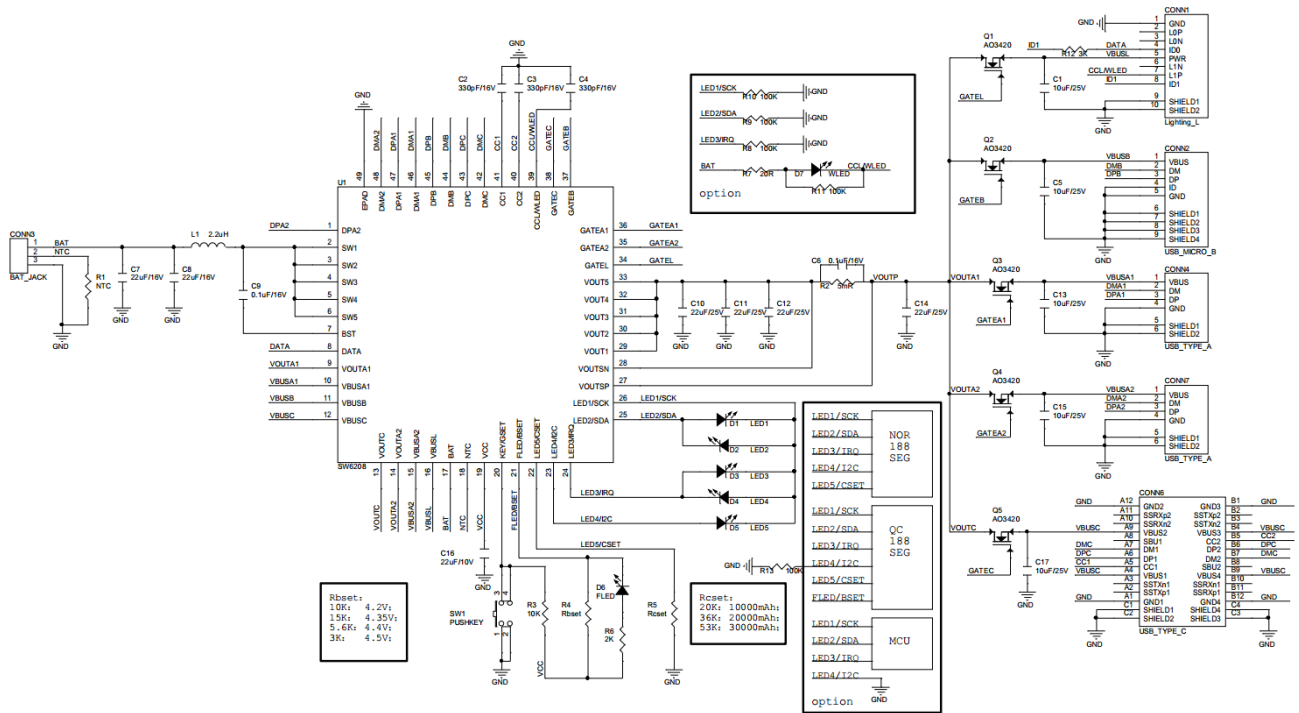
Write Timing:

Slave address : 0x3C

Register address: 0xB0

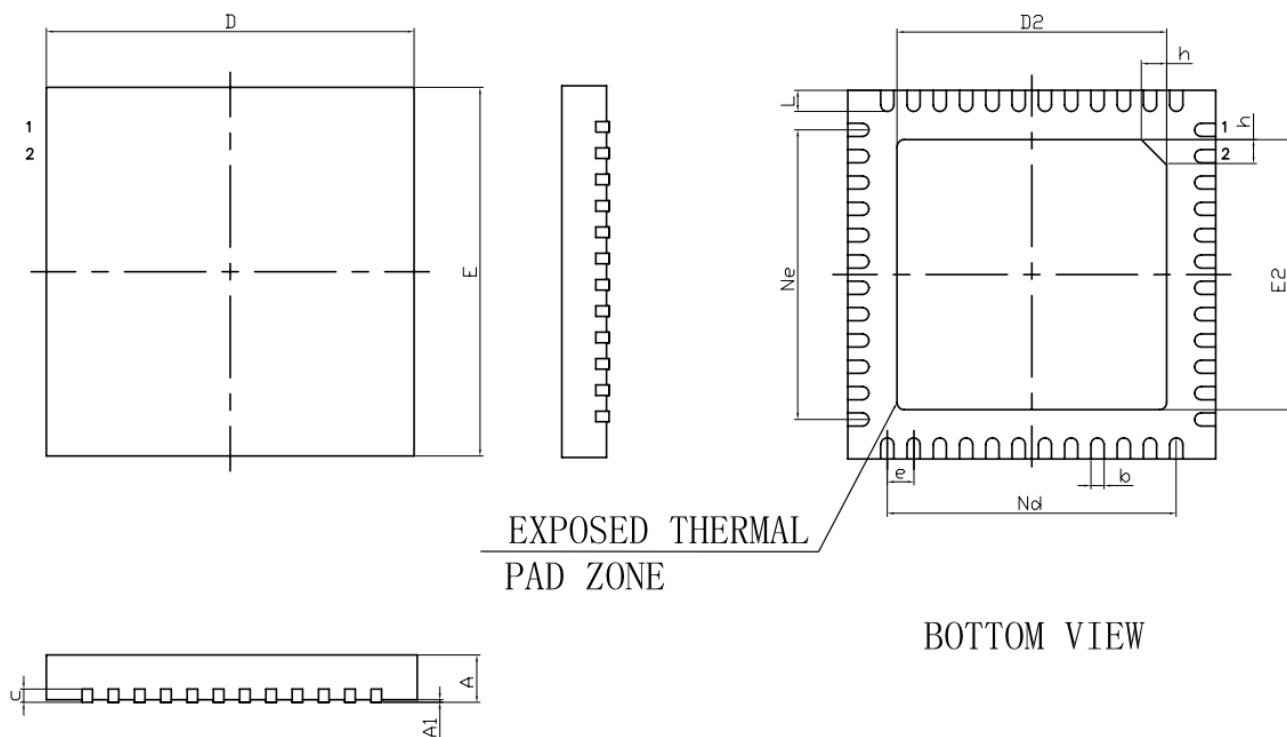


10. Typical Application Circuits



11. Mechanical and Packaging

11.1. Package Summary



11.2. Package Outline and Dimensions

Symbol	Dimension in Millimeters		
	MIN	NOM	MAX
A	0.70	0.75	0.80
A1	-	0.02	0.05
b	0.15	0.20	0.25
c	0.18	0.20	0.23
D	5.90	6.00	6.10
D2	4.10	4.20	4.30
e	0.40BSC		
Ne	4.40BSC		
Nd	4.40BSC		
E	5.90	6.00	6.10
E2	4.10	4.20	4.30
L	0.35	0.40	0.45
h	0.30	0.35	0.40

12. Revision History

- V1.0 Initial version.
- V1.1 Modify some electrical characteristics.
- V1.2 Modify description of NTC protection in boost mode.
- V1.3 Modify company logo.
- V1.4 Modify PD PDO parameters.
- V1.5 Update document template.

Disclaimer

Zhuhai iSmartWare Technology Co., Ltd. (hereinafter referred to as "iSmartWare") may modify or update the products, services and this document provided at any time without prior notice. Customers should obtain the latest relevant information before placing an order and confirm that such information is complete and up-to-date.

The information contained in this document is for your convenience only, and iSmartWare makes no representations or warranties, express or implied, written or oral, statutory or otherwise, for such information, including but not limited to the purpose, characteristics, conditions of use, merchantability of the product. etc. iSmartWare does not assume any responsibility for this information and the consequences caused by the unreasonable use of this information.

iSmartWare assumes no obligation for application assistance or customer product design. Customers are solely responsible for their products and applications using iSmartWare. Customers should provide sufficient design and operation security verification, and guarantee that when integrating iSmartWare products into any application, they will not infringe third-party intellectual property rights, and iSmartWare will not be responsible for any infringements.

When resale of iSmartWare products, if there are differences or false components compared with the product parameters and their statements, all express or implied authorizations of iSmartWare related products will be automatically lost, and this is unfair and fraudulent. For business behavior, iSmartWare reserves the right to protect its rights in all legal ways. iSmartWare assumes no responsibility or liability for any such misrepresentation.

This document is only allowed to be reproduced without any tampering with the content and with relevant authorizations, conditions, restrictions and statements, otherwise iSmartWare has the right to pursue its legal responsibility. iSmartWare assumes no responsibility or liability for such tampered documents. Reproduction of information involving third parties is subject to additional restrictions.